

Generalized Problem Statement

Evaluate the generalized limit for constants $a, b, c \in \mathbb{R}$:

$$L = \lim_{x \rightarrow 0^+} \frac{1}{x} \int_0^x (1 + a \sin(bt))^{\frac{c}{t}} dt$$

Detailed Solution

To evaluate the generalized limit L , we follow the same principles of calculus as the specific case, utilizing the Fundamental Theorem of Calculus and L'Hôpital's Rule.

Step 1: Apply L'Hôpital's Rule

Rewrite the limit as a quotient to identify the indeterminate form:

$$L = \lim_{x \rightarrow 0^+} \frac{\int_0^x (1 + a \sin(bt))^{\frac{c}{t}} dt}{x}$$

As $x \rightarrow 0^+$, the definite integral from 0 to x approaches 0, and the denominator x approaches 0. This yields an indeterminate form of type $\frac{0}{0}$, permitting the use of L'Hôpital's Rule. We differentiate the numerator and the denominator with respect to x .

By the Fundamental Theorem of Calculus, the derivative of the numerator is the integrand evaluated at x :

$$\frac{d}{dx} \left(\int_0^x (1 + a \sin(bt))^{\frac{c}{t}} dt \right) = (1 + a \sin(bx))^{\frac{c}{x}}$$

The derivative of the denominator is 1. Therefore, applying L'Hôpital's Rule yields:

$$L = \lim_{x \rightarrow 0^+} \frac{(1 + a \sin(bx))^{\frac{c}{x}}}{1} = \lim_{x \rightarrow 0^+} (1 + a \sin(bx))^{\frac{c}{x}}$$

Step 2: Evaluate the Exponential Limit

We are now evaluating a limit of the form 1^∞ . Let this limit be y :

$$y = \lim_{x \rightarrow 0^+} (1 + a \sin(bx))^{\frac{c}{x}}$$

Take the natural logarithm of both sides to bring the exponent down:

$$\ln(y) = \lim_{x \rightarrow 0^+} \ln \left((1 + a \sin(bx))^{\frac{c}{x}} \right) = \lim_{x \rightarrow 0^+} \frac{c \ln(1 + a \sin(bx))}{x}$$

This produces another $\frac{0}{0}$ indeterminate form. We apply L'Hôpital's Rule a second time. Differentiating the numerator with respect to x requires the chain rule:

$$\frac{d}{dx}[c \ln(1 + a \sin(bx))] = c \cdot \frac{1}{1 + a \sin(bx)} \cdot (ab \cos(bx)) = \frac{abc \cos(bx)}{1 + a \sin(bx)}$$

The derivative of the denominator x is 1. Substituting these back into our limit:

$$\ln(y) = \lim_{x \rightarrow 0^+} \frac{\frac{abc \cos(bx)}{1 + a \sin(bx)}}{1} = \lim_{x \rightarrow 0^+} \frac{abc \cos(bx)}{1 + a \sin(bx)}$$

Now, we can directly substitute $x = 0$:

$$\ln(y) = \frac{abc \cos(0)}{1 + a \sin(0)} = \frac{abc(1)}{1 + 0} = abc$$

Step 3: Final Calculation

Since we have found that $\ln(y) = abc$, we exponentiate both sides to solve for y :

$$y = e^{abc}$$

Thus, the generalized limit L evaluates to e^{abc} . (Note: In the original problem, $a = 1, b = 2, c = 1$, which gives $e^{(1)(2)(1)} = e^2$).

Final Answer:

$$\mathbf{e^{abc}}$$